Letters

Andronov–Hopf Oscillator-Based Phase-Locked Loop for Three-Phase Grid Connected Inverters

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***Abstract*—This letter proposes a virtual oscillator phase-locked loop (VO-PLL) that integrates an Andronov-Hopf oscillator to improve synchronization in distorted grid. VO-PLL replaces the conventional proportional integral (PI) and resettable-integrator in traditional PLL, with a nonlinear Hopf oscillator known for its ability to sustain orbitally stable limit cycle oscillations at specific frequency. Inherent nonlinear dynamics of oscillator results in fast dynamic performance, harmonic rejection, and improved frequency tracking. Experimental tests show strong evidence for the effectiveness of the proposed VO-PLL, demonstrating its potential to enhance grid synchronization and imitate the operation of a higher order PLL under grid disturbances.**

***Index Terms*—Phase-locked loops (PLL), virtual oscillator (VO).**

I. Introduction

THE three-phase synchronous reference frame phase-locked loop (SRF-PLL) is widely used in grid connected inverters (GCIs) for grid synchronization, however, its performance is deteriorated in distorted grid. Reducing bandwidth of SRF-PLL improves disturbance rejection but slows detection speed, which may be unsuitable for applications requiring fast response [1]. To address this issue, many advance PLLs have been proposed in the literature. These PLLs incorporate specific filtering stages, within the control loop (in-loop filters or in-LF) or before the input (prefilters). Commonly used in-LFs, such as notch filters and moving average filters, mitigate high-frequency oscillations in the estimated frequency or phase angle [2], [3]. Similarly, prefiltered PLLs, such as the decoupled double synchronous reference frame PLL [4], multireference frame-based PLL [5], dual-second-order generalized-integrator-based PLL [6], and multiple complex-coefficient-filter-based PLL [7], focus on the rejection of double-frequency disturbances caused by negative-sequence component. Aforementioned PLLs are capable of addressing low-order harmonics and dc offset. However, the dynamics of the filtering stages is often disregarded. Golestan et al. [8] addressed this oversight and concluded that for effective disturbance rejection, the open-loop bandwidth of the PLLs should be tuned lower than the lowest harmonic frequency. The discussion highlights that increasing the type of a PLLs system reduces its stability margin and dynamic response. In contrast, lowering the PLL type enhances both dynamic performance and stability margin; however, PLL fails to achieve zero steady-state phase error under grid distortions and frequency changes. Virtual oscillator control (VOC) makes inverters mimic the dynamics of nonlinear limit-cycle oscillators. It replaces traditional stationary and rotatory reference frame current controllers and PLLs with nonlinear oscillators. In VOC-based control, GCIs replicate the dynamics of nonlinear oscillators, such as Duffing–Zeeman Oscillator and Van der Pol Oscillator [9], [10]. This control strategy, when implemented in inverters, enables them to generate self-sustained and stable oscillations. These characteristics facilitate synchronization and power sharing among inverters without the need for communication, and also support the regulation of frequency and voltage [10]. Although VOC shows superior dynamic performance as compared to conventional droop control-based applications, however, due to the absence of inner-loop controller, the current limiting feature of inverter is difficult to achieve (without additional control modifications) in VOC, which poses significant safety issue [11].

As discussed above, the nonlinear oscillators used in VOC techniques enhance the GCI synchronization with a demerit of current limiting, due to the absence of inner current controller. Therefore, the proposed PLL integrates a nonlinear oscillator with a conventional DQ controller, improving GCI synchronization in weak grid conditions while maintaining current limiting. The motivations and innovations of this letter are summarized as follows: 1) this letter proposes a novel virtual oscillator-based PLL (VO-PLL) that leverages the Andronov-Hopf oscillator's nonlinear dynamics for fast and robust synchronization under grid disturbances; 2) the proposed VO-PLL demonstrates effective disturbance rejection capabilities, including frequency

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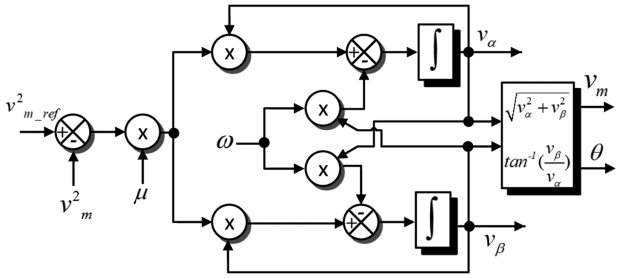

Fig. 1. Block diagram of Andronov-Hopf oscillator.

ramps, phase jumps, and grid harmonics, and exhibits performance similar to a Type-III PLL. Experimental results with GCI demonstrate that the proposed VO-PLL achieves superior synchronization and disturbance rejection compared with Type-II and Type-III SRF-PLLs, particularly under distorted and extremely weak grid conditions. Moreover, VO-PLL's compatibility for real-time closed-loop control of grid-tied converters is validated through experimental tests, including harmonics, transient phase jumps, grid sequence reversal, frequency ramps, high line impedance in weak grids, and, lastly, steady-state grid connected operation.

The rest of this letter is organized as follows. Section II presents the Andronov–Hopf oscillator background. Section III gives detailed VO-PLL design and mathematical model. Section IV analyzes the disturbance rejection in VO-PLL based on describing function method. Section V gives the parameter design for the VO-PLL, and Section VI presents experimental results where proposed VO-PLL is used in GCI to compare with Type-II and Type-III PLLs, and also, active power injection test is performed with real grid to test the closed loop performance. Finally, Section VII conclude this letter.

II. Overview of Andronov–Hopf Oscillator

Fig. 1 shows the block diagram of Andronov–Hopf oscillator, and it is a nonlinear dynamic system that produces stable limit cycle oscillations and gives output as two orthogonal components at a given frequency ω. Unlike second order generalized integrators (SOGI), the structure is composed of linear differential equations and typically works as a bandpass filter, whereas the Hopf oscillator has quadratic nonlinearity, which can sustain oscillations even without any input. Its model can be described by state-space equations of orthogonal components of the output voltage vector, v_α and v_β, in the stationary reference frame

$$\frac{d}{dt}\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \mu\left(v_{m_{\text{ref}}}^2 - \|\mathbf{v}\|^2\right)\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} + \omega\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} \quad (1)$$

$$\|\mathbf{v}\| = v_m = \sqrt{v_\alpha^2 + v_\beta^2} \quad (2)$$

where μ is the bifurcation parameter, which is the amplitude's growth rate parameter, v_m is the instantaneous voltage magnitude, $v_{m_{\text{ref}}}$ is the desired reference amplitude, and ω is the instantaneous angular frequency.

The amplitude dynamics of the oscillator are regulated by the term $\mu(v_{m_{\text{ref}}}^2 - v_m^2)$ in (1). If $v_m > v_{m_M}$, the oscillation amplitude will decrease, driving the system back to stability. In contrast, if $v_m < v_{m_{\text{ref}}}$, the amplitude will increase until it reaches the $v_{m_{\text{ref}}}$. This self-regulation mechanism ensures that the system maintains a stable limit cycle at the reference magnitude $v_{m_{\text{ref}}}$. The phase angle θ of the output voltage is derived from the $\alpha\beta$ components as

$$\theta = \tan^{-1}\left(\frac{v_\beta}{v_\alpha}\right) \quad (3)$$

The output angle of the oscillator with respect to its orthogonal components is represented by the phase angle θ. The Hopf oscillator can control the output phase angle by changing two parameters, i.e., the angular frequency ω and the magnitude $v_{m_{\text{ref}}}$. In a closed loop, these parameters effectively synchronize GCIs with the grid. The next section discusses the integration of the Hopf oscillator into the proposed VO-PLL.

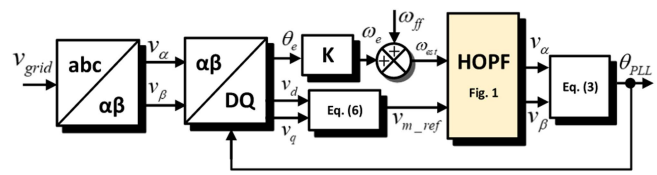

Fig. 2. Proposed virtual oscillator VO-PLL block diagram.

III. Proposed VO-PLL Algorithm

Fig. 2 shows the proposed VO-PLL block diagram. VO-PLL is a modified SRF-PLL, where PI controller and integrator of an SRF-PLL are replaced with a proportional gain and Hopf oscillator, respectively. Hopf oscillator in VO-PLL works as a synchronization block, which generates output as v_α and v_β components and phase angle, which is then given to abc-dq transformation as a feedback to complete a closed loop. The mathematical model of VO-PLL can be derived from an abc-dq transformation where distorted three-phase grid voltages are transformed into the rotatory reference frame

$$\begin{aligned}\begin{bmatrix} v_d \\ v_q \end{bmatrix} &= \mathbf{T}(\theta_{\text{PLL}})\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} \\ &= \mathbf{T}(\theta_{\text{PLL}})\left(\begin{bmatrix} v_{\alpha_1} \\ v_{\beta_1} \end{bmatrix} + \sum_{h\in\{5,7,11\}} \Re\left\{e^{jh\omega t}\begin{bmatrix} v_{\alpha_h} \\ v_{\beta_h} \end{bmatrix}\right\}\right) \\ &= \underbrace{\mathbf{T}(\theta_{\text{PLL}})\begin{bmatrix} v_{\alpha_1} \\ v_{\beta_1} \end{bmatrix}}_{\text{Fundamental}} + \underbrace{\sum_{h\in\{5,7,11\}} \mathbf{T}(\theta_{\text{PLL}})\Re\left\{e^{jh\omega t}\begin{bmatrix} v_{\alpha_h} \\ v_{\beta_h} \end{bmatrix}\right\}}_{\text{Harmonic Components}}\end{aligned} \quad (4)$$

$$\mathbf{T}(\theta_{\text{PLL}}) = \begin{bmatrix} \cos(\theta_{\text{PLL}}) & \sin(\theta_{\text{PLL}}) \\ -\sin(\theta_{\text{PLL}}) & \cos(\theta_{\text{PLL}}) \end{bmatrix} = \Re\left\{e^{-j\theta_{\text{PLL}}}\right\} \quad (5)$$

Equation (4) shows that the dq components contain the fundamental frequency component and contributions from the 5th, 7th, and 11th harmonics. The magnitude of the voltage vector

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in the rotatory frame is given as

$$v_{m_{\text{ref}}} = \|\mathbf{v}_d + \mathbf{v}_q\|_2 = \sqrt{\sum_{i\in\{d,q\}} \left| v_{i_1} + \sum_{h\in\{5,7,11\}} v_{i_h} \right|^2} \tag{6}$$

where v_{d_h} and v_{q_h} are the contributions from each harmonic to the v_d and v_q components, whereas the phase error θ_e is given as follows:

$$\theta_e \approx v_q \approx \sin(\theta_{\text{grid}} - \theta_{\text{PLL}}) \tag{7}$$

The phase error θ_e, as described in (7), is sampled in an ADC with a discrete time. Therefore, it can be approximated as a time difference in the phase error θ_e over the sampling period T_s to give the frequency error ω_{error}. This approximation captures the frequency dynamics by treating the phase error difference as a discrete approximation of the frequency deviation as

$$\omega_{\text{est}} = \omega_{\text{ff}} + K \cdot \theta_e = \omega_{\text{ff}} + K \cdot \underbrace{\frac{\theta_e[n] - \theta_e[n-1]}{T_s}}_{\omega_{\text{error}}} \tag{8}$$

where K is the proportional gain and ω_{ff} is the feedforward fundamental frequency. In the proposed VO-PLL, the steady-state error can be eliminated by utilizing only the proportional gain. This is due to Hopf oscillator's internal integral action. $v_{m_{\text{ref}}}$ from (6) and ω_{est} from (8) are given as an input to Hopf oscillator as defined by (1). The dynamics of the Hopf oscillator after substitution is given as

$$\frac{dv_\alpha}{dt} = \mu\left(v_{m_{\text{ref}}}^2 - v_m^2\right) v_\alpha - \omega_{\text{est}} v_\beta \tag{9}$$

$$\frac{dv_\beta}{dt} = \mu\left(v_{m_{\text{ref}}}^2 - v_m^2\right) v_\beta + \omega_{\text{est}} v_\alpha \tag{10}$$

Expressions (9) and (10) show the dynamics of the Hopf oscillator, which includes the extracted grid frequency and amplitude. Thus, the limit cycle generated by the oscillator is synchronized with the grid voltages. The next section investigates the effects of the harmonics present in grid voltage that do not affect the synchronization of the proposed VO-PLL.

IV. Describing Function Analysis and Harmonic Rejection in the Hopf Oscillator-Based VO-PLL

The Hopf oscillator is a set of nonlinear differential equations that cannot be analyzed accurately using linear control theory techniques, such as Bode and Nyquist plots. Therefore, the nonlinear theory describing the function method is applied. The describing function tool is used in nonlinear control systems to approximate the response of a nonlinearity to sinusoidal inputs. To apply the describing function in the Hopf oscillator, the average dynamics of the system for a specific frequency are considered, i.e., fundamental and 5th, 7th, and 11th harmonics. The describing function $N(A)$ is a complex function that represents the amplitude ratio and phase shift between the input (i.e., the grid voltage) and the output (i.e., the PLL's response). The

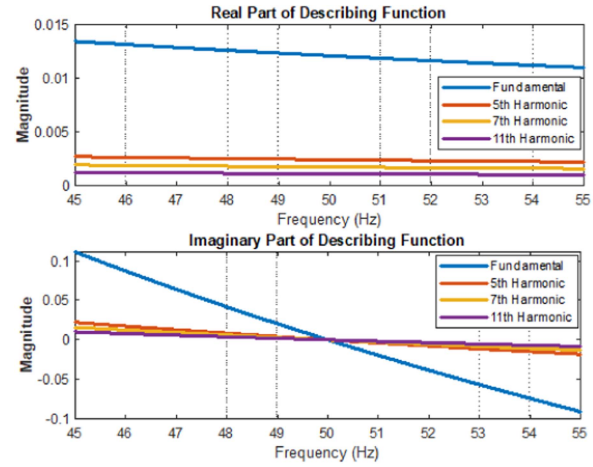

Fig. 3. Magnitude plot of the Hopf oscillator describing function.

differential equations (9) and (10) are transformed as

$$\frac{d}{dt}\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \mu\left(v_{m_{\text{ref}}}^2 - A^2\right)\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} + \begin{bmatrix} -\omega_{\text{est}} & -\omega \\ \omega & \omega_{\text{est}} \end{bmatrix}\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} \tag{11}$$

where A is the steady-state amplitude of the oscillator. After differentiating, (11) becomes

$$\frac{d\mathbf{v}}{dt} = \mu(v_{m_{\text{ref}}}^2 - A^2)\mathbf{v} + \begin{bmatrix} 0 & -\omega_{\text{est}} \\ \omega_{\text{est}} & 0 \end{bmatrix}\mathbf{v} \tag{12}$$

Equating the coefficients of $\sin(\omega t)$ and $\cos(\omega t)$ in these equations leads to the following relationships:

$$\omega = \omega_{\text{est}}$$
$$A = v_{m_{\text{ref}}} \tag{13}$$

Expression in (13) indicates that in the steady state, the oscillator frequency tracks the input frequency, and the oscillator amplitude converges to the reference magnitude $v_{m_{\text{ref}}}$. After further derivation, the final describing function $N(A)$ is

$$N_1(A) = \left[\frac{\mu\left(v_{m_{\text{ref}}}^2 - A_1^2\right)}{\omega} + j\frac{(\omega_{\text{est}} - \omega)}{\omega}\right] \tag{14}$$

$$N_h(A) = \left[\frac{\mu\left(v_{m_{\text{ref}}}^2 - A_h^2\right)}{h\omega}\right] + j\left[\frac{(\omega_{\text{est}} - h\omega)}{h\omega}\right] \tag{15}$$

The describing functions in (14) and (15) describe the real and imaginary components of $N(A)$, showing the gain and phase shift characteristics of the VO-PLL's response to the fundamental and harmonic frequencies. $N(A)$ for each harmonic is plotted in Fig. 3 to observe the tracking of harmonics versus the fundamental frequency. The real part, i.e., Fig. 3 (Top), shows the gain, which indicates the highest magnitude at fundamental frequency with lower gain at other harmonics. Attenuation ratio of harmonics versus fundamental is 10, i.e., -20 dB. Moreover, the imaginary part, i.e., Fig. 3 (Bottom) shows the phase shift, where the fundamental crosses zero at 50 Hz. It shows that the fundamental frequency after synchronization will have zero

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phase shift, and any change in the frequency will be tracked dynamically with the rapid change in the phase shift. The plot shows the VO-PLL's ability to prioritize the fundamental frequency and suppress harmonic distortion. Next section discusses detailed parameter design for VO-PLL.

V. VO-PLL Parameter Design

The design of VO-PLL consists of only two parameters μ and K. The parameter μ influences the convergence settling time T_s to the stable limit cycle. The Hopf osicllator in (1) can be approximated to its amplitude dynamics, the model for amplitude A can be given by

$$\frac{dA}{dt} \approx \mu \left(v_{m_{\text{ref}}}^2 - A^2\right) A \tag{16}$$

By linearizing (16) at $v_{m_{\text{ref}}}{}^*$ and substituting $A = v_{m_{\text{ref}}} - \Delta A$ into the amplitude dynamics

$$\frac{d\Delta A}{dt} \approx -2\mu v_{m_{\text{ref}}}^2 \Delta A \tag{17}$$

where $\Delta A = v_{m_{\text{ref}}} - A$, (17) gives an exponential decay model with a rate constant of $2\mu v_{m_{\text{ref}}}^2$

$$\Delta A(t) = \Delta A(0) e^{-2\mu v_{m_{\text{ref}}}^2 t} \tag{18}$$

From linear control theory, for a system that decays exponentially, the approximate 2% settling time T_s is

$$T_s \approx \frac{4}{\text{decay constant}} = \frac{4}{2\mu v_{m_{\text{ref}}}^2} = \frac{2}{\mu v_{m_{\text{ref}}}^2} \tag{19}$$

$$\mu \approx \frac{2}{T_s v_{m_{\text{ref}}}^2} \tag{20}$$

Thus, μ can be calculated from (20). The parameter K corresponds to system dynamic response to disturbances. To determine the gain K based on the desired settling time T_s, the system's time dynamics are approximated as a first order. For first-order systems, the settling time to reach 98% of the reference is

$$T_s \approx \frac{4}{K} \hookrightarrow K \approx \frac{4}{T_s} \tag{21}$$

Thus, by specifying the desired settling time T_s in (20) and (21), parameters K and μ can be calculated to achieve the desired transient response. For the VO-PLL design, T_s is assumed around 2 mS and $v_{m_{\text{ref}}}$ as 1 for normalized amplitude, and by substituting in (20) and (21), values of K and μ are calculated as 1000 and 2000, respectively. These parameters are used in simulation and experiments where VO-PLL shows actual settling time of 5 mS, which is slightly higher than the theoretical value of 2 mS due to linearization and first-order approximation errors.

VI. Experimental Results

The experimental validation of the proposed VO-PLL is conducted on a GCI setup (two-level voltage source inverter) with a dSPACE (MicroLabBox), as shown in Fig. 4, where the sampling frequency for discrete implementation is 10 kHz. Cinergia GE15 grid emulator is used to emulate a distorted grid and waveforms are captured on the YOKOGAWA DLM4038 oscilloscope. VO-PLL is compared with the conventional Type-II general PI-based SRF-PLL and general Type-III second-order in-LF-based SRF-PLL, as reviewed in [2]. For a fair comparison, the bandwidth of Type-II and Type-III PLLs is tuned to track the grid angle within one and two cycles, respectively. The test grid contains 0.05 per unit (p.u.) of each 5th, 7th, and 11th harmonics. Moreover, active power injection from GCI into the real grid is also conducted to test the closed loop performance of VO-PLL with a three-phase DQ-current controller.

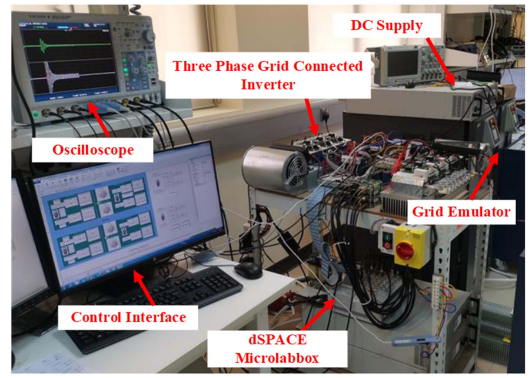

Fig. 4. dSPACE (MicroLabBox) GCI experimental setup.

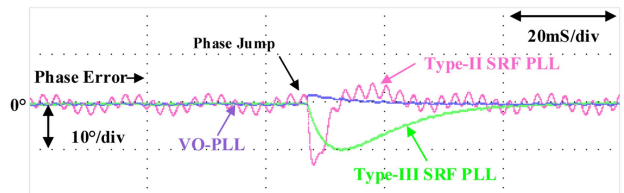

Fig. 5. Phase jump test.

A. Phase Jump Test

The phase jump test is shown in Fig. 5, where 10° phase shift is introduced under distortion to show the transient performance. The VO-PLL showed peak overshoot in phase-error of 2.5° with a settling time of approximately 5 mS, whereas Type-II and Type-III SRF-PLLs showed significant overshoot of 12.2° and 10°, respectively. Moreover, the settling time for Type-II and Type-III SRF-PLLs are 20 and 35 mS, respectively.

B. Frequency Ramp Test

As shown in Fig. 6, the grid frequency ramp is introduced from 50 to 60 Hz at a rate of 5 Hz/s. Fig. 6 (Top) shows the overall effect of the frequency where Type-II SRF-PLL and Type-III PLL exhibit the steady-state error during the ramp. Fig. 6 (Bottom) depicts the zoomed-in image, where Type-II PLL shows the steady-state error of 11.5° and Type-III PLL shows approximately 2° during the frequency ramp. The VO-PLL demonstrated excellent phase-tracking capability, showing negligible phase error of 0.2° during the frequency ramp, highlighting its robustness in tracking frequency changes.

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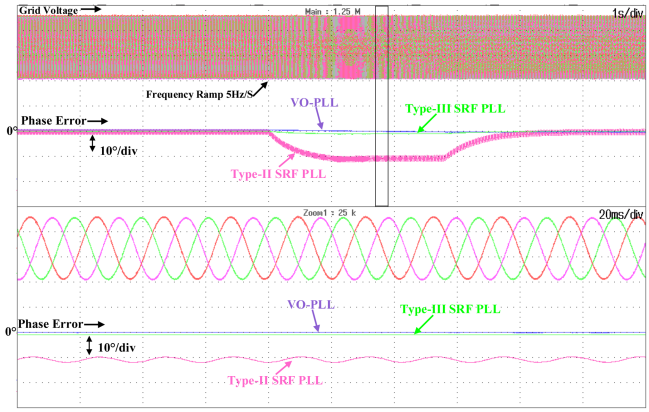

Fig. 6. Frequency ramp test from 50 to 60 Hz, i.e., 5 Hz/s.

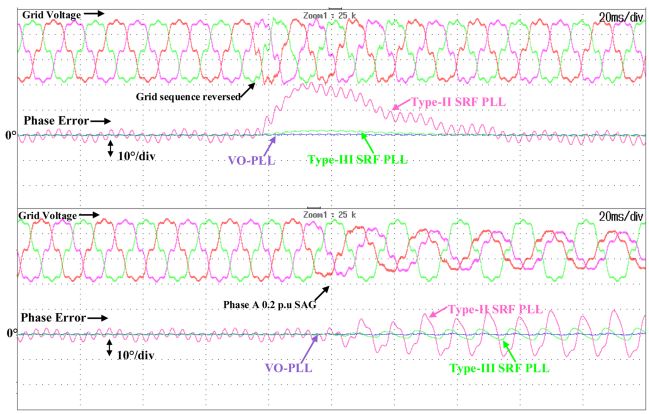

Fig. 7. Phase sequence reverse test (Top) and 0.8 p.u. single phase sag test (Bottom).

C. Phase Sequence Reversal and Voltage Sag Test With Harmonics

Fig. 7 presents the phase sequence reversal and single-phase sag tests under distorted grid conditions to showcase the transient performance. In the phase sequence reversal test, Fig. 6 (Top), the VO-PLL successfully tracked the grid frequency with approximately 0.1° phase error, whereas the Type-II and Type-III SRF-PLLs show the transient peak error of 21° and 2°, respectively. It is important to note that during the steady-state operation under distorted grid conditions Type-III PLL and VO-PLL shows similar phase error. This demonstrates that the VO-PLL has steady-state operation similar to Type-III PLL; however, the transient response is better than both Type-II and Type-III PLLs. In the voltage sag test Fig. 7(Bottom), a 0.2 p.u. sag was introduced in Phase A. The VO-PLL maintained stable tracking with negligible phase error, demonstrating its robustness under voltage sags and unbalanced operation. In contrast, the Type-II SRF-PLL exhibits steady-state oscillations in phase error with a peak of 9.5°, whereas the Type-III SRF-PLL showed better performance than the Type-II with a peak error of 2.5°. The proposed VO-PLL shows a peak error of 0.3°, VO-PLL shows the rejection of the grid amplitude changes once the stable limit cycle is achieved.

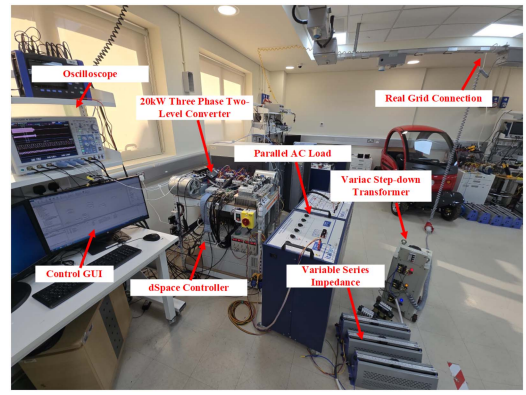

Fig. 8. Experimental setup for a closed-loop grid connected operation test.

TABLE I
SYSTEM PARAMETERS FOR GCI EXPERIMENT

Parameter	Value
Rated Power	20 kW
Switching and Sampling Frequency	10 kHz
DC Link Voltage	300 V
Inverter Side Inductor (L_i)	2.5 mH
Grid Side Inductor (L_g)	0.8 mH
Grid Series Impedance (X_s)	$0.8mH + 5\,\Omega$
Grid Voltage	100 V
Grid Frequency	50 Hz

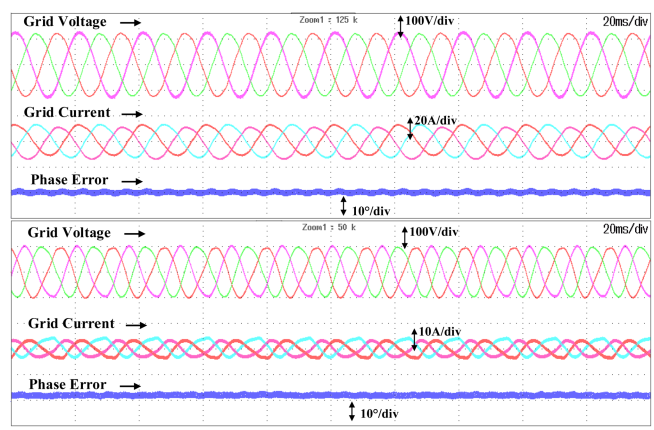

Fig. 9. Closed-loop GCI test using dq-SRF controller and VO-PLL, strong grid (Top) and weak grid (Bottom).

D. Closed-Loop Test With DQ-Controller in GCI

To demonstrate the effectiveness of the proposed method in real-time closed-loop control, a GCI using a dq SRF current controller was implemented with VO-PLL. The experimental setup for the GCI closed-loop test with real grid is shown in Fig. 8, and system parameters are summarized in Table I. A variable series impedance X_s is added after the GCI's *LCL* filter to emulate weak grid conditions. The series impedance includes an inductance of 0.8 mH and an adjustable resistance of 5 Ω. A slider allows the tuning of X_s to switch between strong and weak grid scenarios. In addition, a parallel ac load is connected to absorb inrush currents during grid connection. A Variac transformer reduces the grid voltage from 220 to 100 V L-L for compatibility with the inverter. Fig. 9 (Top)

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TABLE II
COMPARISON BETWEEN VO-PLL, TYPE-II, AND TYPE-III PLLS

Method	Settling Time	Steady-State Error	Disturbance Rejection
Type-II SRF-PLL [2]	20 ms	High	Low
Type-III SRF-PLL [2]	35 ms	Low	Moderate
Proposed VO-PLL	**5 ms**	**Low**	**High**

shows the results under strong grid conditions. In this case, the series impedance X_s is set to zero, leaving only the *LCL* filter impedance. The waveforms show that the grid is being injected with 8 A rms current. The phase error during the grid-connected operation remains negligible; however, it is distorted with a small switching noise and EMI generated during a closed-loop operation with a grid. The proposed VO-PLL effectively maintained synchronization with the grid, as evidenced by the sinusoidal current and stable power output.

Fig. 9 (Bottom) illustrates the grid voltage, current, and phase error during a weak grid scenario. In this case, the series impedance X_s is set to include 0.8 mH inductance and 5 Ω resistance. The grid voltage is further reduced to 60 V L-L using the Variac transformer. During the closed-loop operation, the waveforms show noticeable distortion in grid voltage and current due to the high line impedance, which increases harmonics and unbalances. Despite these distortions, the VO-PLL maintained stable synchronization, with minimal phase error and harmonic distortion. This demonstrates the proposed method's capability to operate effectively under high line impedance, making it suitable for weak grid environments.

VII. CONCLUSION

Performance comparison summary is listed in Table II. The proposed VO-PLL shows improved dynamic response and disturbance rejection compared with Type-II and Type-III SRF-PLL. The Hopf oscillator's inherent fast nonlinear dynamics allow for fast and precise synchronization with enhanced disturbance rejection. Describing function analysis also supports the proposed VO-PLL disturbance rejection capability. VO-PLL requires only two parameters to be tuned based on settling time only. Experimental results under the weak grid condition validate the performance of the proposed VO-PLL, showcasing its potential for enhancing GCIs synchronization.

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